Graph Machine Learning

Some Preliminaries, Random Walk on Graphs, PageRank

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Linear Algebra Preliminaries

Vector Norms

- ☞ A function f that measures the size of a vector is called a norm.
- ☞ A vector norm has to satisfy the following:

$$f(\mathbf{x}) = 0 \rightarrow \mathbf{x} = \mathbf{0}$$

$$f(\mathbf{x}+\mathbf{y}) \leq f(\mathbf{x}) + f(\mathbf{y}) \text{ triangle inequality}$$

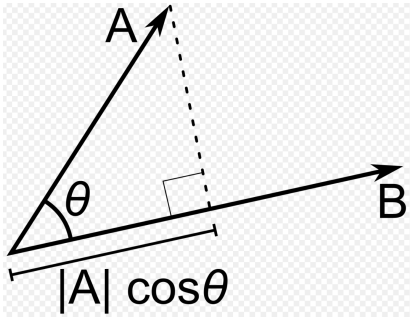
$$\forall \alpha \in \mathbb{R} \; f(\alpha \mathbf{x}) = |\alpha| f(\mathbf{x})$$

- ☞ The general form of vector norm: $||\mathbf{x}||_p = \sqrt[p]{\sum_i |x_i|^p}$
- ☞ The most commonly used is euclidian norm $||\mathbf{x}||_2 = \sqrt{\sum_i |x_i|^2}$

Dot and Cross Vector Product

☞ **A dot product or scalar product of two vectors** is a scalar quantity

- algebraic formula: $||\mathbf{x}||_2||\mathbf{y}||_2 cos(\theta)$ or $\sum_i \mathbf{x}_i \mathbf{y}_i$
- how much two vectors points in same directions. The dot product of two orthogonal vectors is 0.



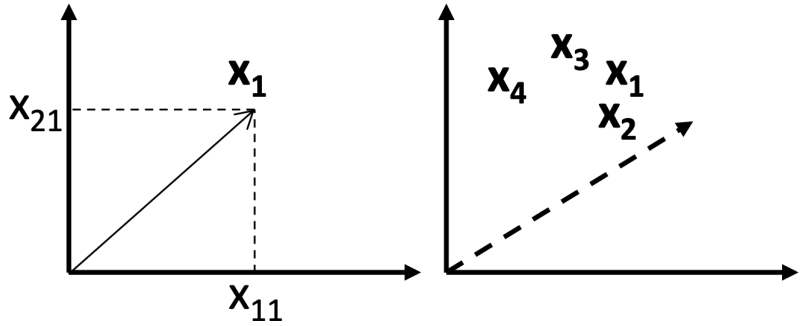
- **A cross or vector product of two vectors** is a vector orthogonal to to both the vectors.
  - algebraic formula: $||\mathbf{x}||_2||\mathbf{y}||_2 sin(\theta)\mathbf{1}$ or use cofactor matrix
  - how much two vectors points in different directions. The cross product of two linear vectors is **0**.

Vectors and Matrices

- Let $\mathbf{x}_i$ $i \in [1..m]$ be m vectors of n elements:
  $\mathbf{x}_1 = \begin{bmatrix} x_{11} & x_{21} & \dots & x_{n1} \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} x_{12} & x_{22} & \dots & x_{n2} \end{bmatrix}$, ... $\mathbf{x}_m = \begin{bmatrix} x_{1m} & x_{2m} & \dots & x_{nm} \end{bmatrix}$
- Matrix representation of size $n \times m$, as a row vector or column vector (transpose)

$$\begin{bmatrix} x_{11} & x_{12} & \dots & x_{1m} \\ x_{21} & x_{22} & \dots & x_{2m} \\ \dots & \dots & \dots & \dots \\ x_{n1} & x_{n2} & \dots & x_{nm} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \dots & \mathbf{x}_m \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \dots \\ \mathbf{x}_m \end{bmatrix}^{\mathsf{T}}$$

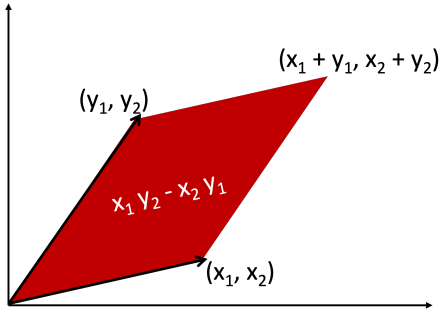
- Each $\mathbf{x}_{i \in [1..m]}$ is n-dimensional point



Matrix Determinant

☞ Let a 2×2 square matrix $\begin{bmatrix} \mathbf{x} & \mathbf{y} \end{bmatrix} = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \end{bmatrix}$, its determinant is a scalar value which characterises some properties of what $(\mathbf{x}, \mathbf{y})$ represents.

$$\det \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \end{bmatrix} \text{ or } \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} = x_1 y_2 - y_1 x_2 = \|\mathbf{x}\| \cdot \|\mathbf{y}\| \cdot \sin\theta$$



✓ The absolute value of the determinant is equal to the area of the parallelogram defined by $\mathbf{x}, \mathbf{y}$

✓ The determinant is = 0 if and only if $\mathbf{x}, \mathbf{y}$ are co-linear (line-parallelogram)

☞ In the case of 3×3 matrix $(\mathbf{x}, \mathbf{y}, \mathbf{z})$. The absolute value of the determinant is the volume of the parallelogram defined by $\mathbf{x}, \mathbf{y}, \mathbf{z}$ = 0 if they are on the same plan (plat parallelogram)

$$\det \begin{bmatrix} \mathbf{x}, \mathbf{y}, \mathbf{z} \end{bmatrix} = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = x_1 \begin{vmatrix} y_2 & z_2 \\ y_3 & z_3 \end{vmatrix} - y_1 \begin{vmatrix} x_2 & z_2 \\ x_3 & z_3 \end{vmatrix} + z_1 \begin{vmatrix} x_2 & y_2 \\ x_3 & y_3 \end{vmatrix}$$

☞ Generalisation to vectorial space of dimension n on $\mathbb{R}$ where the vectors define a parallelotope.

Matrix Determinant

- ☞ Many uses of matrix determinant for ex.:
  - ✓ Coefficients in a system of linear equations $\mathbf{Y} = \mathbf{AX}$, can be used to solve these equations (Cramer's rule), other methods computationally much more efficient.
  - ✓ Characteristic polynomial of $\mathbf{X}$, whose roots are the eigenvalues.
- ☞ Some properties:

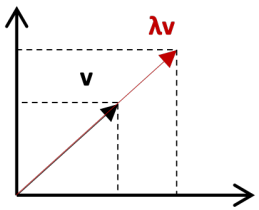
$$\left|\mathbf{I}\right| = 1 \text{ where } \mathbf{I} \text{ identity matrix (1 on the diagonal and 0 elsewhere);}$$

$$\text{If } \left|\mathbf{X}\right| = 0 \ \mathbf{X} \text{ is a singular matrix;}$$

$$\left|\mathbf{XY}\right| = \left|\mathbf{X}\right|\left|\mathbf{Y}\right| \ ; \ \left|\mathbf{X}^{\mathsf{T}}\right| = \left|\mathbf{X}\right| \ ; \ \left|\mathbf{A}^{\mathbf{n}}\right| = \left|\mathbf{A}\right|^{n}$$

Eigenvectors and eigenvalues

- An eigenvector (or characteristic vector) of a linear transformation T is a nonzero vector that changes at most by a scalar factor λ (eigenvalue): $T(\mathbf{v}) = \lambda\mathbf{v}$.



- Eigenvectors and eigenvalues provide valuable insights, through linear transformations, on the structure of data with a high number of features.
- If we apply to vectors a given T: scaling, shearing ($(x, y) \to (x + 2y, y)$) or rotation, some vectors change their direction, and others do not. The vectors that do not change their direction (or rotate to the opposite direction) are the eigenvectors, "characteristic" vectors of the transformation, the eigenvalue is the scaling factor of the transformation (possibly a negative or complex number).
- There is a direct correspondence between $n \times n$ square matrices and linear transformations of an n-dimensional vector space into itself, given any vector space basis.
  The T representation of $\mathbf{A}$ $n \times n$ matrix: $\mathbf{Av} = \lambda\mathbf{v}$.

Eigenvectors and eigenvalues

- ☞ Finding all eigenvalues: Solving a polynomial function of λ called the characteristic polynomial of $\mathbf{A}$: $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = 0$ has a nonzero solution $\mathbf{v}$ iff $\left|\mathbf{A} - \lambda\mathbf{I}\right| = 0$
- ☞ Example: $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$; $|\mathbf{A} - \lambda\mathbf{I}| = \begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = 3 - 4\lambda + \lambda^2 = (\lambda - 1)(\lambda - 3) = 0$

$$v_1 + v_2 = 0 \text{ if } \lambda_1 = 1 \text{ and } -v_1 + v_2 = 0 \text{ if } \lambda_2 = 3$$

  The eigenvectors define *eigenSpace*(λ_1) and *eigenSpace*(λ_2) ($t \in \mathbb{R}^*$): $\mathbf{v}_{\lambda_1} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} t, \quad \mathbf{v}_{\lambda_2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} t$

- ☞ An infinity of collinear eigenvectors (cross product null) for λ_1 and also for λ_2.
  Digression: Strictly speaking, we talk about the eigenvector associated with one given eigenvalue

  Note that not all matrices have eigenvalues.

Eigenvectors and eigenvalues

- ☞ Algebraic multiplicity t_i of eigenvalue λ_i

$$(\lambda - \lambda_1)^{t_1}(\lambda - \lambda_2)^{t_2}\ldots(\lambda - \lambda_k)^{t_k} = 0$$

  $\sum_i t_i = n$ where $t_i \in \mathbb{N}^*$ satisfying the algebraic multiplicity of λ_i
- ☞ **A** can have at most n distinct eigenvalues (complex or real).
- ☞ If eigenvalues of **A** are distinct values, then the corresponding eigenvectors are linearly independent (non collinear).

Eigenvectors and eigenvalues

- ☞ Example: $\begin{bmatrix} -1 & 1 & 0 \\ -4 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$

  $(\lambda-1)^2(\lambda-2)=0$ has two eigenvalues: $\lambda_1=1$ and $\lambda_2=-2$ with algebraic multiplicities 2 and 1, resp. and $\mathbf{v}_{\lambda_1}$ is defined by $-2v_1+v_2=0$ and $v_1+v_3=0$

  the *eigenSpace*(λ_1) is $\mathbf{v}_{\lambda_1}=\begin{bmatrix} -t \\ -2t \\ t \end{bmatrix}$ $\forall t\in\mathbb{R}^*$. This set has dimension 1.

- ☞ Example: $\begin{bmatrix} 4 & 6 & 0 \\ -3 & -5 & 0 \\ -3 & -6 & 1 \end{bmatrix}$ $(\lambda-1)^2(\lambda+2)=0$

  $\begin{bmatrix} 2t \\ -t \\ u \end{bmatrix}$ $\forall(t,u)\in\mathbb{R}^*$ *eigenSpace*$(\lambda=1)$ dimension 2. $\begin{bmatrix} -t \\ t \\ t \end{bmatrix}$ $\forall t\in\mathbb{R}^*$ *eigenSpace*$(\lambda=-2)$ dimension 1.

- ☞ The dimension of *eigenSpace*(λ_i) is the geometric multiplicity of λ_i whose the value is at most its algebraic multiplicity.

Eigenvectors and eigenvalues

What about:

- ☞ $\begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$
- ☞ $\begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{bmatrix}$
- ☞ $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$
- ☞ $\begin{bmatrix} 0 & -1 & -1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$

Graph Representation using Matrices

☞ $\mathcal{G}(E, V)$ with n vertices and m edges can be encoded using:

- ✓ Adjacency Matrix $\mathbf{A}(n \times n)$, $n = |V|$

$$\mathbf{A}_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \in E, \\ 0 & \text{otherwise} \end{cases}$$

  Symmetric matrix if $\mathcal{G}$ is an undirected (without loops)

- ✓ Adjacency list $\mathbf{L}$ each vertex holds a list of its neighbours

$$\forall v_i \in V,\ \mathbf{L}_i = \{v_j | (v_i, v_j) \in E\}$$

  If $\mathcal{G}$ is directed the choice of the direction depends on analytic needs

- ✓ Incidence matrix $\mathbf{B}$, $n \times m$

$$\mathbf{B}_{ij} = \begin{cases} 1 & \text{if } e_j = (v_i, v_k) \in E, \\ 0 & \text{otherwise} \end{cases}$$

Adjacency Matrix Power

Using the adjacency matrix $\mathbf{A}$, the number of r- length paths:

$$\mathbf{A}^r = \mathbf{A} x \mathbf{A}^{r-1} \forall r > 1$$

$\mathbf{A}^r_{ij} = \sum_k \mathbf{A}^{r-1}_{ik} \mathbf{A}_{kj}$, the number of r-length paths between nodes i and j
Computing $\mathbf{A}^r$ is $\mathcal{O}(rn^3)$: requires r multiplications, each one $\mathcal{O}(n^3)$.
If r is pair $\mathbf{A}^r = (\mathbf{A}^{\frac{r}{2}})^2$, otherwise $\mathbf{A}^r = \mathbf{A}(\mathbf{A}^{\frac{r-1}{2}})^2$. Computing $\mathbf{A}^r$ requires $\mathcal{O}(log_2(r))n^3$ multiplications.
The distance between v_i and v_j is the smallest d such that $\mathbf{A}^d_{ij} \neq 0$
Useful to compute shortest paths, triangles, ...

First Perron-Frobenius Theory

- ☞ Perron-Frobenius vector for non-negative matrices leads to the characterization of non-negative primary eigenvectors, such as stationary distributions of Markov chains and Google's PageRank.
- → Note that positive or non negative matrix iff $\mathbf{A}_{ij} > 0$ resp. $\mathbf{A}_{ij} \geq 0$ is different from positive definite or semi-positive definite matrix
- ☞ Perron theorem for positive matrix $\mathbf{A}_{ij} > 0$, it exists a real, positive eigenvalue (Perron root or value)
  - ✓ $\exists \lambda^* > 0, \mathbf{v}^* > \mathbf{0}, ||\mathbf{v}^*||_2 = 1$ s.t $\mathbf{A}\mathbf{v}^* = \lambda^* \mathbf{v}^*$ right column eigenvector
    $\exists \lambda^* > 0, \mathbf{w} > \mathbf{0}, ||\mathbf{w}||_2 = 1$ s.t $\mathbf{w}\mathbf{A} = \lambda^* \mathbf{w}$ left row eigenvector
  - ✓ $\forall \lambda$ other eigenvalue of $\mathbf{A}, |\lambda| < \lambda^*$, dominant eigenvalue (the largest absolute value)
  - ✓ λ^* is simple (multiplicity 1) and $\mathbf{v}^*$ is positive and unique (up to rescaling).
    Such eigenvectors are called Perron vectors.
- ☞ Frobenius extension for non negative matrix (next): $\mathbf{v}^*$ are non negative and the uniqueness of $\mathbf{v}^*$ is not guaranteed

First Perron-Frobenius Theory

- **Definition: Irreducible**
  Connected graph $\mathcal{G}(V, E) \Leftrightarrow$ for any $1 \leq i, j \leq |V|, \exists k \in \mathbb{N}^*, \mathbf{A}_{ij}^k > 0$
- **Definition: Primitivity (Aperiodicity)**
  strengths this condition to k-path length, *i.e.* every pair of nodes are connected by a path of length $k \Rightarrow \lambda^*$ is unique and $\lambda^* = max|\lambda|$
  - ✓ $\mathbf{A}$ is connected (irreducible) and $\mathbf{A}_{ii} > 0$ for some i is sufficient for primitivity but not necessary
- **Definition: Primitive + Non-negative**
  - ✓ $\exists \lambda^* > 0, \mathbf{v}^* > \mathbf{0}, ||\mathbf{v}^*||_2 = 1$ s.t $\mathbf{A}\mathbf{v}^* = \lambda^* \mathbf{v}^*$ right column eigenvector
    $\exists \mathbf{w} > \mathbf{0}, ||\mathbf{w}||_2 = 1$ s.t $\mathbf{w}\mathbf{A} = \lambda^* \mathbf{w}$ left row eigenvector
  - ✓ $\forall \lambda$ other eigenvalue of $\mathbf{A}, |\lambda| < \lambda^*$, dominant eigenvalue (the largest absolute value)
  - ✓ $\mathbf{v}^*$ is unique (up to rescaling).

Random walks on graphs

- A random walk on a graph $\mathcal{G}(E, V)$ is a random process that starts from some vertex v_i, and repeatedly moves to a neighbor v_j chosen uniformly at random. ξ_t is a random variable describing the position of a random walk after t steps.

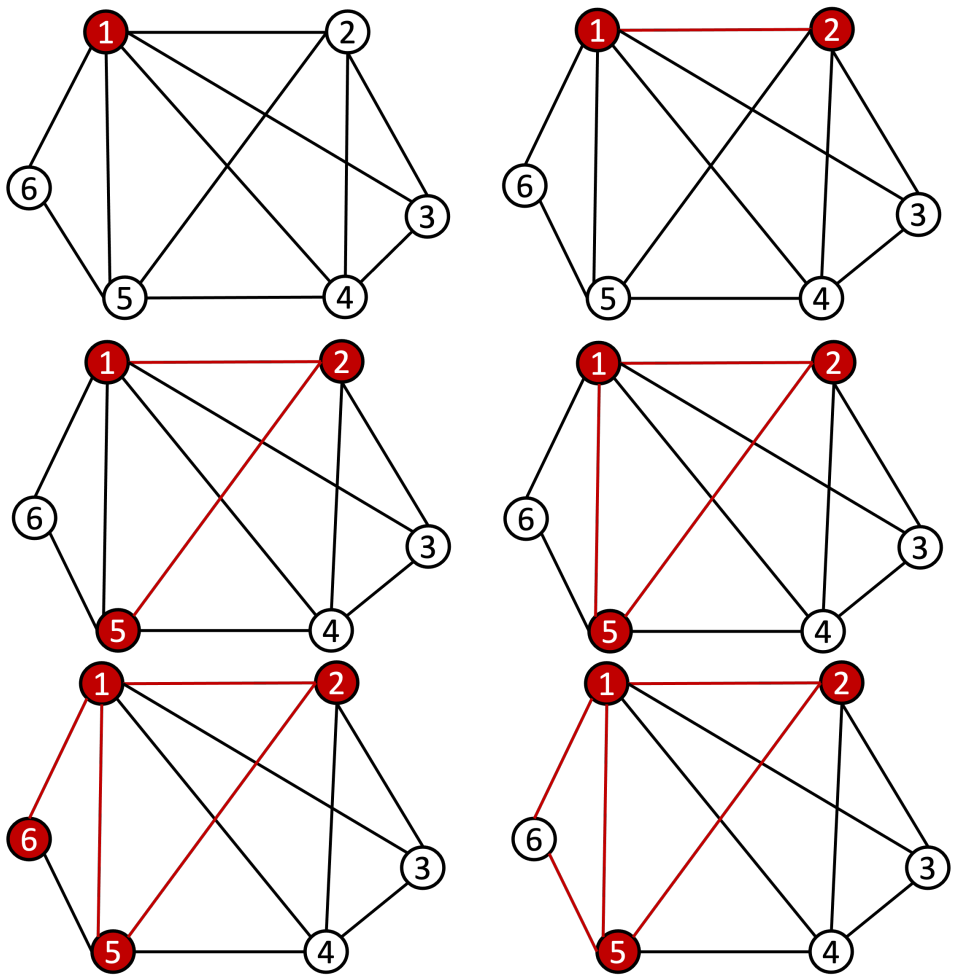
$$\mathbf{P}_{ij} = P(\xi_{t+1} = j | \xi_t = i)$$

- The sequence can be regarded as a category of Markov chain (discrete time stochastic process) where the position ξ_0 is the initial state according to the distribution P^0 and next state depends only on the current state. The t steps transition probability is:

$$\mathbf{P}^t_{ij} = P(\xi_t = j | \xi_0 = i)$$

- Some examples: path traced by a molecule in a liquid or a gas (Brownian motion) , the price of a fluctuating stock, the financial status of a gambler, ... The term random walk was first introduced by Karl Pearson in 1905.

Random walks on graphs: Example



Random walks on graphs

- Consider a random walk with $\mathbf{P}_{ij} = P(\xi_{t+1} = j | \xi_t = i) \geq 0$, thus $\mathbf{P}$ is a row-stochastic or row-Markov matrix:

$$\sum_j \mathbf{P}_{ij} = 1, \mathbf{P} \times \mathbf{1} = \mathbf{1} \in \mathbb{R}^n \text{ right eigenvector} > 0$$

- The value of any eigenvalue of a stochastic matrix cannot exceed 1. A general property of eigenvalues is that any eigenvalue λ_i is bounded by any induced matrix norm $||\mathbf{P}||$
  The norm of a stochastic matrix induced by vector row norms is $||\mathbf{P}||_\infty = Max_i(||\mathbf{P}_i||)$
  $\rightarrow \lambda = 1$ is the dominant value.
- From Perron-Frobenius theorem for non-negative matrices, we know:
  - ✓ $\mathbf{v}^* = \mathbf{1}$ is a right Perron eigenvector of $\mathbf{P}$
  - ✓ $|\lambda| \leq \lambda^* = 1$, is a Perron eigenvalue
  - ✓ $\exists$ left Perron row eigenvector $\pi\mathbf{P} = \pi$
  - ✓ $\mathbf{P}$ is primitive $\Rightarrow \pi$ is unique

Random walks on graphs: Stationary distribution

- Let be π^t the row vector giving the probability distribution of ξ_t. π_i^t be the probability that a walk is on v_i at t.

$$\pi^{t+1} = \pi^t \mathbf{P}$$
$$\pi^{t+1} = \pi^0 \mathbf{P}^t$$

$$lim_{t\to\infty}\pi^{t+1} = lim_{t\to\infty}\pi^t \mathbf{P}$$

$$\lambda\pi = \pi\mathbf{P} \text{ with } \lambda = 1$$

- This means if we take powers of $\mathbf{P}$, *i. e.*, $\mathbf{P}^k$, all rows will converge to the stationary distribution π if primitivity holds.

Random walks on graphs

- ☞ What does "random" mean? If the walk is on i at t, the single step transition probability refers to the uniform probability that the random walk moves to j at $t+1$.

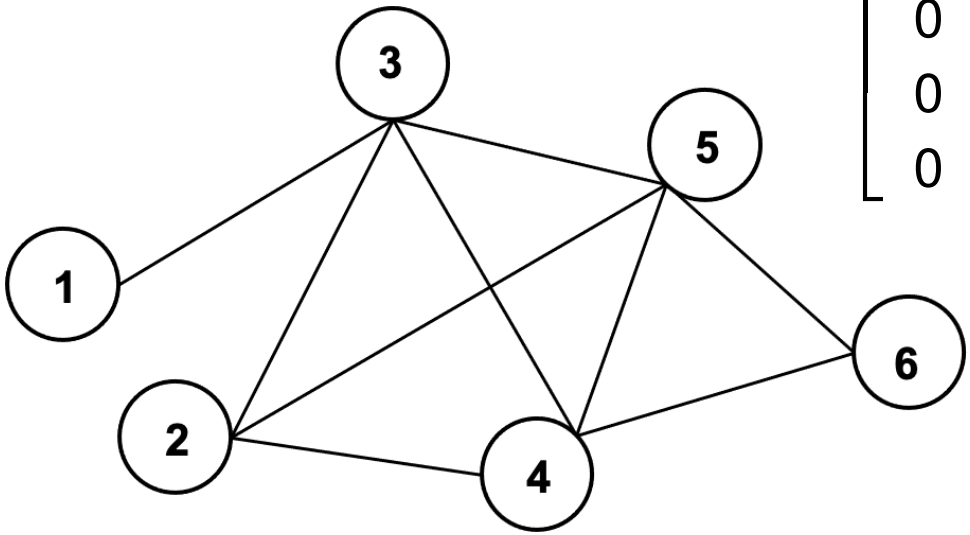
$$\mathbf{P}_{ij} = P(\xi_{t+1} = j | \xi_t = i) = \begin{cases} \frac{1}{d_i} & \forall v_i, v_j \in V \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

$$\mathbf{P}_{ij} = \frac{\mathbf{A}_{ij}}{\sum_{v_j \in V} \mathbf{A}_{ij}} = \frac{\mathbf{A}_{ij}}{d_i} = \mathbf{D}_{ii}^{-1} \mathbf{A}_{ij}$$

- ☞ The random sequence of vertices $\xi_0, \xi_1, \ldots \xi_t, \xi_{t+1}, \ldots$ visited on $\mathcal{G}(E, V)$ is a Markov chain with state space V and matrix transition probabilities

$$\mathbf{P} = \mathbf{D}^{-1} \mathbf{A}$$

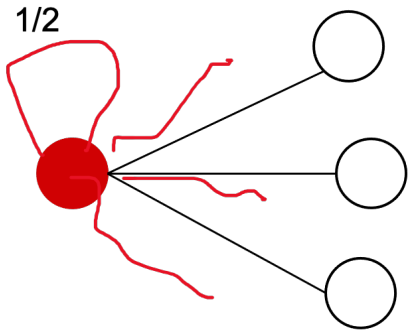
Random walks on graphs



$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & 1/3 & 1/3 & 0 \\ 1/4 & 1/4 & 0 & 1/4 & 1/4 & 0 \\ 0 & 1/4 & 1/4 & 0 & 1/4 & 1/4 \\ 0 & 1/4 & 1/4 & 1/4 & 0 & 1/4 \\ 0 & 0 & 0 & 1/2 & 1/2 & 0 \end{bmatrix}$$

Random walks on graphs: Lazy random walk

- Lazy random walk: either stay on the current node with the probability $\frac{1}{2}$ or walk on a neighbor



$$\mathbf{P}_j^{t+1} = \frac{1}{2}\mathbf{P}_j^t + \frac{1}{2}\sum_i \frac{\mathbf{A}_{ij}}{d_i}\mathbf{P}_i^t$$

- Matrix Form

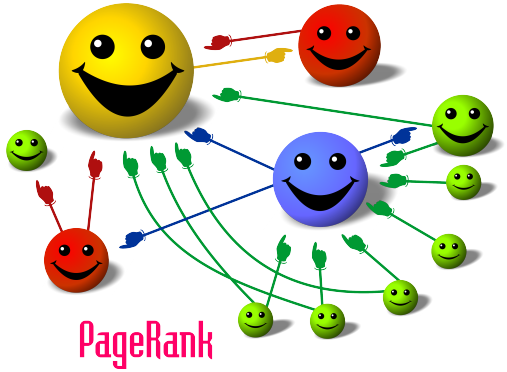
$$\mathbf{P}^{t+1} = \frac{1}{2}\mathbf{P}^t(\mathbf{I} + \mathbf{D}^{-1}\mathbf{A})$$

- Convervges to the same stationary distribution

$$(2\lambda - 1)\pi = \pi(\mathbf{D}^{-1}\mathbf{A})$$

$\lambda = 1$ is the eingenvalue

PageRank



- Formalize the problem in the case of directed graph by taking into account only out going edges
- The web is very heterogeneous by its nature, and certainly huge, we do not expect its graph to be connected. The solution of Page and Brin: fix a positive constant p between 0 and 1, called the damping factor (a typical value for p is 0.15). Define the Page Rank matrix

$$\mathbf{P}_g = (1-p)\mathbf{P} + p\mathbf{B} \text{ where } \mathbf{B} = \frac{1}{n}\begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & 1 & \dots & 1 \\ \dots & \dots & \dots & \dots \\ 1 & 1 & \dots & 1 \end{bmatrix}$$

  Most of the time, a surfer will follow the outgoing links and move on to one of the neighbors. A smaller percentage of the time, the surfer will dump the current page and choose arbitrarily a different page from the web. The damping factor p reflects the probability that the surfer quits the current page and jump to a new one .

- Prove that $\mathbf{P}_g$ remains stochastic.